

Generative AI for Business Expense Management

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Abstract—The system uses a small language model , making it elegant for employees to submit their expense reports. With the elimination of shifting between applications, this system therefore makes the process quicker and saves more time and effort in processing. This expense bot assists the user when navigating through submitting details such as the type of expense and the receipt images, eliminating complex workflows. In this regard, the SLM automatically categorizes expenses by separating travel, accommodation, and meal types. This intelligent categorization minimizes errors on the part of the system user and saves the burden of manual classification by the users themselves. Moreover, it is equipped with Optical Character Recognition, wherein the system will scan the uploaded receipts to extract text from there. This means that all the details captured are only those on the names and totals with the vendors, to then be processed by the SLM, to the right categorization of them.

Upon the submission of an expense report, the system automatically lets the approver selected by the system know, thus eradicating manual routing of the same. Such notifications are sent, hasten up the approval, allowing managers to review and approve submissions quickly without unnecessary delays. This new system streamlines expense reporting by automating categorization and speeding up the approval workflow. In addition, it empowers employees to submit their claims easily while accelerating and accurately processing the managers' approval process. Generally, the system minimizes data entry through the manual process, eliminates errors, and increases the productivity of staff and management, allowing them to have time and space for high-value work.

Index Terms—Expense Management, Small Language Models (SLMs), Expense Categorization, Optical Character Recognition (OCR), Automation, Workflow Optimization

I. INTRODUCTION

The current cost management processes in place with most organizations presents several inefficiencies and challenges to employees and managers alike. It is as easy as using a couple of applications, all you need to do is capture the expense report, track their expenses and find what they actually expend those on spending most of it with own hands punching on mobile keyboard and chasing down approvers. Thus, its turnaround time is very long and the figures are inaccurate. These inefficiencies lower the productivity of an organization, as the staff must spend more time on non-core tasks related to administration.

This tackles the issues by presenting an automated expense management system driven by a Small Language Model that has been hyperparameter-tuned. It looks forward to making the process of submitting an expense report less time-consuming and feasible, so that users may submit one without having to

switch between applications too often. As a result, there is less chance of error and the expenditure submission process is completed more quickly.

Therefore, an expense bot that guides a user through the process of entering their expenses forms the core of this system. Important details like the type of expense, the amount paid, the date, and the ability to submit receipt photographs can all be entered. After the image is uploaded, OCR scans it to extract pertinent text and information. The SLM then analyzes this data to determine which classification actually exists. The SLM

automates categorization processes, but it smartly classes up expenses appropriately into categories which include, for example, travel, lodging, and meals. To this extent, automation of categorization reduces the need for hand operations like data entry, consequently lowering the probability of human error by great percentages. Apart from the categorization process, the system collects living cost data that further improves on the correct classification as regards to expenses. Upon the submission of the expense report, the system automatically notifies the approver, reducing a lot of the hassle of manual routing. This eliminates the delays and administrative overhead. Employees obtain their reimbursements fast when the system sends the approvers notifications for approval and processing of expenses as well as routing.

The main advantages of the project are that it automates expense reporting easy, reduces the hours the team needs to input and re-input the same information thus leading to the efficiency of businesses as a whole. First, the system offers an integrated platform doing away with the need to go through multiple programs to file expenditure reports thus cutting time and frustration. Secondly, the expenditure bot and SLM classify the expenses automatically, limiting the quantity of manual input and lowering errors. Third, the automatic notification system helps in shortening the processing period since the report will be checked and approved immediately, thus reducing the time between reimbursement. Finally, the system provides efficiency benefits through the reduction of manual computation, which is one of the core errors contributing to inefficiencies, hence creating a more effective flow for the employees and managers.

Technical scope for this project will be the development of a backend using Python and Flask. The interaction between the user interface and the SLM has been controlled by the backend that is based on a fine-tuned T5 model optimized for

expense categorization. The training data consists of Indian source-picked documents, including invoices and receipts. This system allows the focus on it to be capable enough for processing common types of expenses in that context, and the OCR engine is also trained to give precise text

extraction. However, the system has its drawbacks. It has been developed to cater for expenses reports coming straight from within India, which, naturally enough will impact upon the processing of other regions' documents. There are differences in format and/or language. Thus, this automatic expense management system, with its limitations, is much more advanced than traditional processes. It facilitates faster, accurate, and simpler execution, thereby saving time as compared to the preparation of an expense report, reduces errors, and lifts the burden and sometimes stress from employees and managers in this respect.

II. LITERATURE SURVEY

Kushavaha, Kishan[1] says Optical Character Recognition According to him, it means robotic turn for the textbook information coming out from images translates the published or handwritten documents into something accessible on digital formats that may turn easier to store and process. The quality friction of characters and complexity in handwriting make it difficult for OCR, yet it is an essential part of data entry robotization, document digitization, and hundreds of diligence and operations.

Joshi and Pvt[2] mentioned Tesseract OCR, which referred to open- source textbook recognition machine for more than 100 languages. Out of these, Hindi and Bengali are significant for India. It can digitize published as well as handwritten documents. Compared to AI, it beautifully upgrades the data birth robotization process apart from document digitization across different operations. Exploring book summarization in natural language processing using convolutional T5 and Seq2Seq models.

Lubis et al.[3] discovered that several other variables other than case perceptibility affect the textbook appreciation. With these values, they could fine-tune the Cream criteria by readjusting the subcaste confines, learning rate, and Dropout parameters to reach a best Cream criteria value of 0.8 for Cream- 1, 0.83 for Cream- 2, and 0.8 for Cream- L using the same model.

M.K Ghose, K.B Shah and B.K Thakur[4] presents the current trend and tools that are used for point birth for maximum bracket of the hand written rudiments. There exist many models of point birth defined by several researchers in their individual discussion. It is felt that with the introduction of Euler Number into zoning, speed of classifier prosecution increases along with delicacy since it reduces the dimension of the hunt space as the character set is resolve into three groups.

Patel, A. Patel and D.[5] discusses ways of Optical Character Recognition (OCR) where preprocessing and segmentation of textbook are crucial for obtaining correct birth. The paper also discusses the armature of the open-source Tesseract tool and its performance, which makes it more effective than

the marketable Transym tool in rooting vehicle figures from license plates against various criteria.

Colin Raffel, Noam Shazeer, Adam Roberts, Katherine Lee, Sharan Narang, Michael Matena, Yanqi Zhou, Wei Li, Peter J. Liu [6] provide a notion of transfer literacy in NLP, any textbook- grounded task standardize into textbook- to-textbook and introduce a unified frame. In comparison, the pre-training objects and infrastructures are used to achieve new state-of-the-art results on multiple marks. All the authors have published their dataset, pre-trained models, and law for further NLP exploration.

The Mastropaolo et al. [8] study explores the Text-To-Text Transfer Transformer (T5) in assisting with code-related work within software engineering. Published in the 2021 IEEE/ACM International Conference on Software Engineering, this paper addresses how T5 can support code generation, documentation, and understanding via a text-to-text framework. The outcomes conclude that T5's versatility elevates code-related operations and provides insight to boost efficiency in software development. The contribution of this paper is toward advancing the methodologies of NLP in software engineering and demonstrates the potential capability of T5 toward overcoming many challenges when trying to manage and process code.

Wang et al. [9] (2023) propose a T5-based model for abstractive summarization using semi-supervised learning with consistency loss functions. The study effectively enhances summarization quality by leveraging labeled and unlabeled data. Such a model reflects significant outcomes and proves to be useful in the text summation application, thereby enhancing the results of the natural language processing task in general. Rayhan et al., [10] (2023): Rayhan et al. (2023) discussed advancement in Natural Language Processing that enabled machines to understand human language better. The authors emphasized the most important technologies and methodologies that pushed NLP and how it translates the evolving machine understanding into applications across fields. This work underlines the fast-increasing importance of NLP in current computational systems.

Bharathi[11] Exploring NLP for enterprise applications-the role of NLP in business processes. The paper cites how various NLP technologies, be it sentiment analysis, chatbots, or any automated text processing, are changing the way enterprises function by increasing efficiency, better decision-making, and interaction with customers.

Bozkus, Bisson, and Arsan [12] in 2009 developed an analytical expense management system that was presented at the First International Conference on Networked Digital Technologies. The emphasis of the system is to process tracking and reporting of the expenses of organizations in such a way that it is optimized and done automatically. The idea behind the concept relies on improving the efficiency along with minimizing errors within the operation of the entire expense management workflow through data analysis.

III. PROPOSED METHODOLOGY

This section describes the methodology that was adapted to develop the expense management system. The architecture of the expense management system is shown in figure 1. The architecture consists of five components: Slack Bot, Python Flask Server, T5 Model, the MongoDB Database and Tesseract OCR. The Slack bot serves as an interface using which the user can submit their expense bills and raise an approval request. The data extracted by the OCR and the category predicted by SLM are displayed to the user using the bot. The bot also serves as a medium for the manager to approve or reject the request raised by the user.

The Python Flask Server gets the file uploaded from the user in Slack. The content of the file is read and provided to the T5 Model for classification. The Flask Server also communicates with the MongoDB database to save the request details and update the approval status.

The Tesseract OCR is employed for reading out the text data from the user-uploaded file. The image is processed and the OCR extraction carried out on it. The output of the extraction is taken as input by the T5 model to carry out the classification.

The T5 Model performs classification of invoice. It predicts the invoice category as: stay, restaurant, travel or miscellaneous. This model was trained using the invoice data set from these categories.

The MongoDB database is used to store mapping of users and managers in the organisation and to maintain the request details. The request details such as request id, date of submission, total amount, gst number and the approval status is also stored in the database.

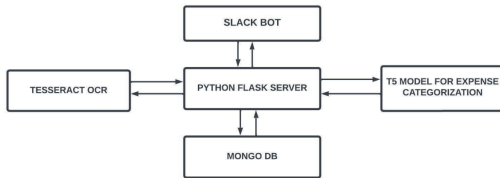


Fig. 1. System Architecture

The project began by identifying the inefficiencies plaguing the current expense management process. We envisioned a solution that streamlined workflows and minimized human error. This led to the concept of a Slack-integrated system powered by a fine-tuned machine learning model.

To build this system, we embarked on a multi-step design process. First, we focused on data acquisition. Pytesseract extracted text from the bill images, and this text data was categorized and stored alongside the corresponding expense types. To enhance the dataset and enhance model performance, we employed GPT-3.5 to generate synthetic data that mimicked the real OCR data structure. The distribution of the dataset is given in Figure 2.

With a robust dataset in place, we turned our attention to model development. We selected a T5 small language

model as the foundation for expense categorization. Through meticulous hyperparameter tuning, we optimized the model's performance. T5 (Text-to-Text Transfer Transformer) is a set of language models designed by Google AI. T5 models are trained on a huge dataset of code and text under a text-to-text paradigm. The T5 models can do the text-based tasks they have been pre-trained to perform. T5 is an encoder-decoder model and translates all NLP issues into a text-to-text type. They can be finetuned for other tasks too. In our project we employ the fine tuned T5 base language model with 220 million parameters, for categorization of expenses. T5 is trained through teacher forcing, i.e., while training we always require an input sequence and a target sequence corresponding to it. T5 is pre-trained on a multi-task mix of unsupervised and supervised tasks. T5 is trained employing a multi-task learning approach, wherein the thought is to group multiple tasks while pre-training the model. The T5 model was fine tuned using the simple T5 wrapper built on top of PyTorch Lightning and Transformers Library.

Following model development, we focused on deployment and user interaction. The fine-tuned model was deployed in a local environment. A Flask-based backend established a bridge between the Slack bot and the Python backend, facilitating seamless user interaction. Users submit expense reports directly within Slack, and the bot efficiently converts the data into OCR text for further processing. The system then proceeds to categorize expenses and generate insights. Based on the extracted city name from the OCR data, the system fetches related city tier information. Such city tier data gives power to the system to offer informative notes on average food and stay prices in that particular place. In conclusion, the fine-tuned SLM steps into the limelight, smartly classifying the bills based on the extracted text data. The last step is reporting and approval. The organized bills and cost-of-living information are gathered and displayed as a cluster in the Slack channel. The approver gets notified in Slack, asking them to review and approve the expense report. This process reduces manual routing requirements and accelerates the approval process. This well-established design process guaranteed the creation of a stable and user-friendly expense management system, enabling employees and managers with a smoother and error-free workflow as depicted in Figure 3.

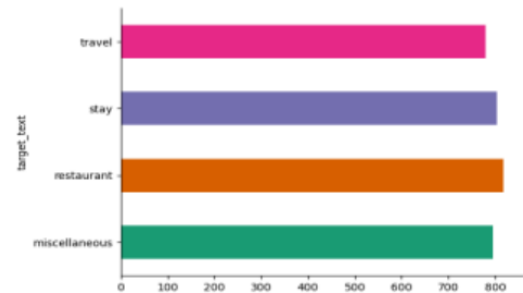


Fig. 2. Distribution of the Dataset

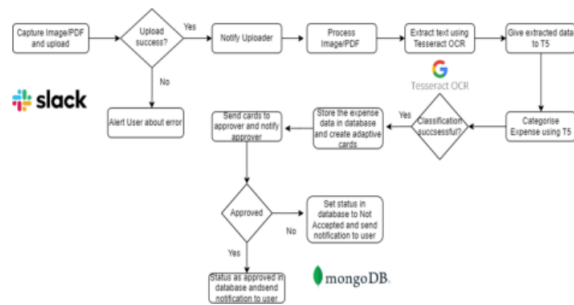


Fig. 3. Workflow

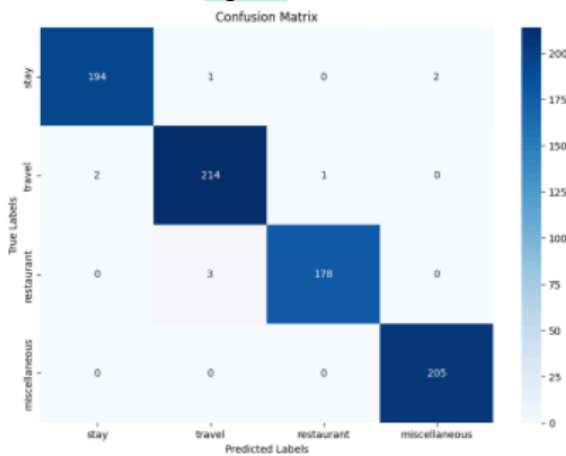


Fig. 4. Confusion Matrix

IV. CONCLUSION

This project led to the development of a revolutionary Slack-integrated expense tracking system. The revolutionary system enables employees to file expense reports directly within the intuitive Slack environment, without the hassle of switching between programs and enabling a more efficient workflow. An inbuilt expense bot asks questions and scans receipt photos, which are then smartly categorized by a well-tuned small language model (SLM). This SLM assures precise expense categorization. In addition, the system also uses Optical Character Recognition (OCR) technology to read text from uploaded receipts, which further enhances the SLM's accuracy in categorization. Upon submission, the system automatically alerts the authorized approver, streamlining the approval process and avoiding manual routing. This innovative method promotes substantial productivity improvements through automating tedious manual data entry, reducing errors, and simplifying the entire expense reporting process for employees as well as managers. Though the initial deployment targets the Indian market. Subsequent releases may expand the geographic coverage by integrating extra data feeds for receipts, invoices, and cost-of-living data in other regions. Further, the precision of the SLM's expense categorization may be refined even more by enlarging the training dataset to cover a greater variety of expense categories and international contexts. This project

provides a good foundation for creating an effectively strong and world-relevant cost management system.

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